

NEERAJ KUMAR

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EDUCATION

Guru Gobind Singh Indraprastha University, New Delhi
Bachelor's Degree - Automation & Robotics - CGPA - 7.00

Nov 2022 – Jul 2026
New Delhi

TECHNICAL SKILLS

Languages: C++17/20, C, Python, JavaScript

Technologies/Frameworks: React, Node.js, Express, PostgreSQL, MongoDB, Redis, WebSockets, Firebase, Docker, Linux/UNIX environments, REST APIs, TCP/IP

Tools: Git, Nginx, GitHub Actions, Vercel, Postman, VS Code

Core Competencies: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, System Design, Distributed Systems

EXPERIENCE

Outlier - AI Code Reviewer & Data Annotator

Mar 2026 – Apr 2026

Freelance

Remote

- Reviewed and evaluated AI-generated code and technical responses for correctness, logic, edge cases, and adherence to task requirements.
- Annotated and ranked model outputs, identifying logical errors, inefficient implementations, and inconsistencies to improve code-generation quality.
- Applied structured evaluation criteria across programming and technical tasks, providing detailed feedback and corrected solutions.

PROJECTS

FixEngine | [GitHub](#) | C++20, CMake, Linux, TCP/IP

May 2026 – Aug 2026

- Built a low-allocation FIX protocol parsing engine in C++20 with a 6-stage pipeline spanning ingestion, tokenization, parsing, validation, formatting, and orchestration.
- Designed bounded, zero-copy message storage using fixed-capacity containers and `std::string_view`; benchmarked core operations at approximately ~ 2 ns/op.
- Implemented **checksum** and **body length validation** with explicit error types, maintaining deterministic, exception-free control flow on the normal processing path.

RedisCLI | [GitHub](#) | C++17, POSIX Sockets, RESP2

Mar 2026 – Apr 2026

- Built a Redis CLI client from scratch in C++17, implementing the **RESP2** protocol **serializer** and **parser**.
- Implemented event-driven **Pub/Sub** mode using `poll()` to multiplex stdin and Redis socket simultaneously - streaming published messages while keeping terminal input responsive via GNU Readline callback API.

Options Pricing Simulator | [GitHub](#) | Python, SciPy, NumPy

Feb 2026 – Mar 2026

- Developed a **Monte Carlo** simulation engine to price **European Options** with configurable simulation runs.
- Validated results against Black-Scholes analytical model, achieving <0.03 pricing deviation at **10,000** simulations.
- Computed option sensitivities : Delta (Δ), Vega (ν), Gamma (Γ), and Theta (Θ) - to support quantitative risk analysis.

Discord-Vault | [GitHub](#) | Python, SHA256

Jul 2026 – Sep 2026

- Developed a secure Discord-based file vault for storing and retrieving encrypted files through a bot interface.
- Implemented **SHA-256** based integrity verification to detect file tampering and ensure data consistency.
- Implemented lossless PNG compression and decompression, reducing image storage overhead while preserving pixel-perfect data integrity.
- Automated file upload, retrieval, and vault management through Discord commands, eliminating the need for a dedicated frontend.

CERTIFICATIONS

- Postman API Fundamentals Student Expert** - [Postman](#)
- Career Essentials in GitHub Professional Certificate** - [LinkedIn Learning](#)
- Software Engineering Job Simulation (JPMorgan Chase & Co.)** - [Forge](#)

ACHIEVEMENTS & EXTRACURRICULARS

- Achieved Max Rating of **1614** on [LeetCode](#); solved **300+** algorithmic problems on LeetCode in C++.
- Achieved Global Rating of **123** on CodeChefs and max rating of **1159** on [Codeforces](#).
- Represented the college Volleyball team in inter-collegiate tournaments and secured **3rd place**.